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Domain: Artificial Intelligence & Machine Learning

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Experiment 3

Lab Tasks



Experiment 3.py

Viva Questions

1. A NumPy array is faster and uses less memory than a Python list because it stores elements of the same data type. NumPy arrays support vectorized operations, meaning we can perform mathematical operations on the entire array without using loops, while Python lists do not support this. NumPy arrays also provide many built-in mathematical functions like sum, mean, and standard deviation which are not available in lists.
2. The shape of an array tells us the number of elements in each dimension of the array. For example, shape (3, 4) means the array has 3 rows and 4 columns. It helps us understand the structure of the data and is important when performing operations like matrix multiplication or reshaping the array.
3. Axis=0 means we sum along the rows, which gives us the sum of each column. Think of it as going downwards vertically. For example, in a 2-D array with 3 rows and 4 columns, axis=0 will return 4 values, each representing the sum of one column.

4. Vectorization is faster because NumPy operations are implemented in C language and run on optimized machine code. Instead of looping through each element one by one in Python, vectorized operations perform calculations on the entire array at once using optimized C code. This is much faster especially for large datasets.
5. A DataFrame is a table-like data structure in Pandas that has labeled rows and columns, similar to an Excel spreadsheet. Unlike a NumPy array which can only store one data type, a DataFrame can store different data types in different columns (text, numbers, dates). DataFrames also provide many useful features like filtering, grouping, and handling missing values which are not available in NumPy arrays.
6. Missing values can be handled by either removing them or filling them with a suitable value. We can use `dropna()` to remove rows with missing values or `fillna()` to replace them with the mean, median, or mode of that column. In Pandas, we use `df.dropna()` to remove missing values and `df.fillna(value)` to replace them with a specific value.

MCQ's

Q1. The shape of a 3-row, 4-column array is:

(b) (3, 4)

Q2. Which library is best for tabular data?

(b) Pandas

Q3. `df.head()` shows the:

(b) first rows

Q4. A missing value in Pandas is shown as:

(b) NaN

Q5. axis=0 in a 2-D array refers to:
(b) columns down

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Date: 06 July, 2026

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